



Cyberster Blue Team Internship

Week 7 Report

Digital Forensics Lab Report

Prepared by: Muhammad Ubaid

Track: Blue Team Operations

Phase: Digital Forensics and Threat Intelligence

Submitted To: Sir Abdullah Zia (SOC Training Instructor)

Date: 2 May 2026

Core Competencies:

**Evidence Acquisition & Integrity | Forensic Imaging (E01) | NTFS
Metadata Analysis (\$MFT) | Data Hiding Detection (ADS) | MACB
Timeline Reconstruction | Automated Forensics (Autopsy)**

Digital Forensics Lab Report

Examiner: Muhammad Ubaid

Location: Jhelum, Punjab, Pakistan

Case ID: 2026-USB-FORENSIC-01

Date: May 2, 2026

1. Days 46 & 47 — Disk Imaging and Hash Verification

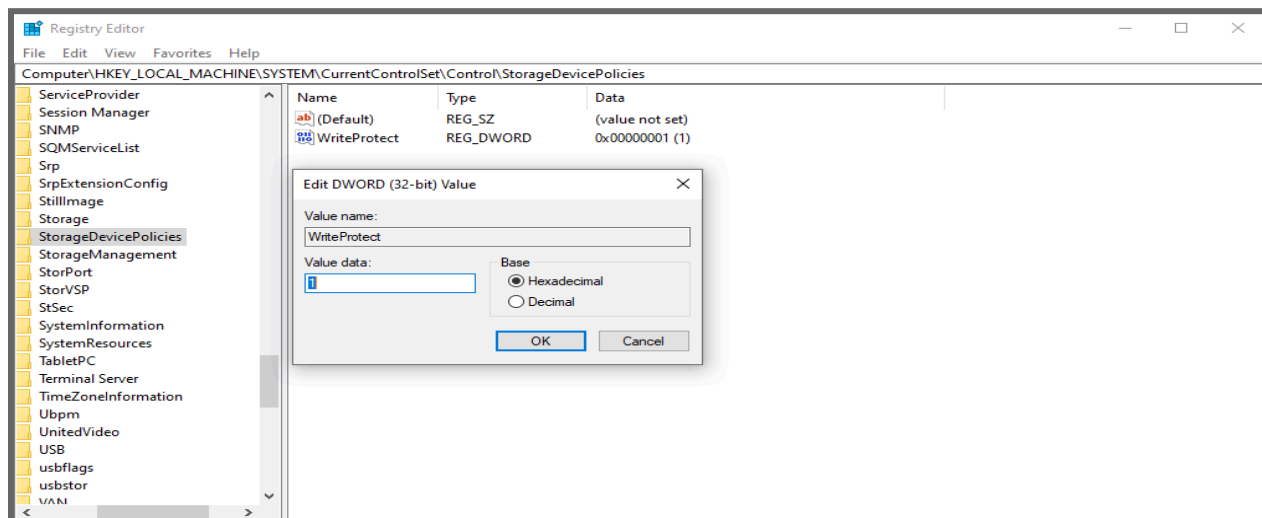
1.1 Executive Summary

The primary goal of this phase was to perform a forensically sound acquisition of a USB storage device. Forensic imaging ensures that an exact, bit-for-bit copy of the physical media is created, including unallocated space and deleted records, which serves as the foundation for the entire investigation.

1.2 Write-Blocking Methodology

In professional forensics, hardware write-blockers are used to prevent the host operating system from modifying the evidence. For this lab, a **Software Write-Block** approach was utilized.

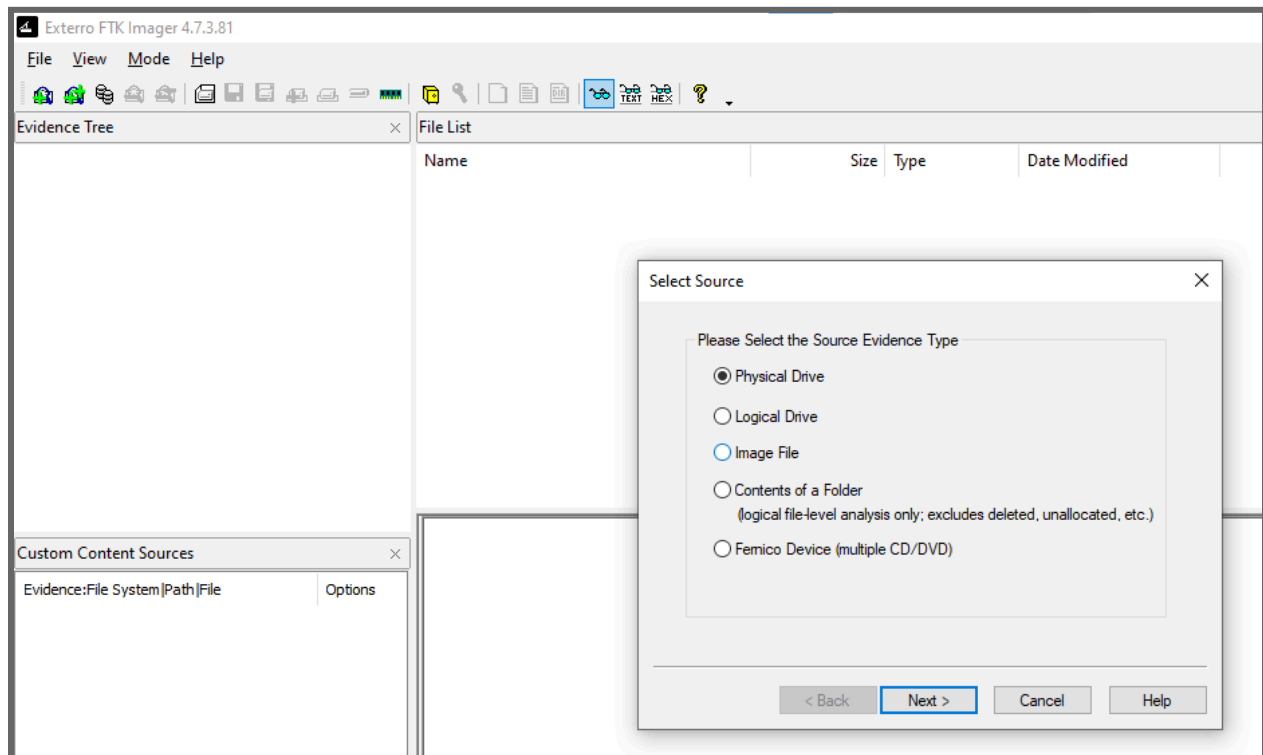
- **Method:** Windows Registry modification (`StorageDevicePolicies\WriteProtect` set to 1).
- **Limitation Note:** While effective for this exercise, software write-blocking is less robust than hardware solutions as it relies on the host OS to honor the restriction.



1.3 Evidence Acquisition (FTK Imager)

The acquisition was performed using **FTK Imager v4.7.3.81**.

- **Source:** Physical Drive (USB Device).
- **Format: E01 (Expert Witness Format).**
- **Why E01?** Unlike RAW images, E01 files support internal compression and embed case metadata (Case Number, Examiner Name, Notes) directly into the file, ensuring a stronger chain of custody.
- **Configuration:** Compression level was set to **6** and fragments were disabled (set to **0**) to maintain a single evidence file.

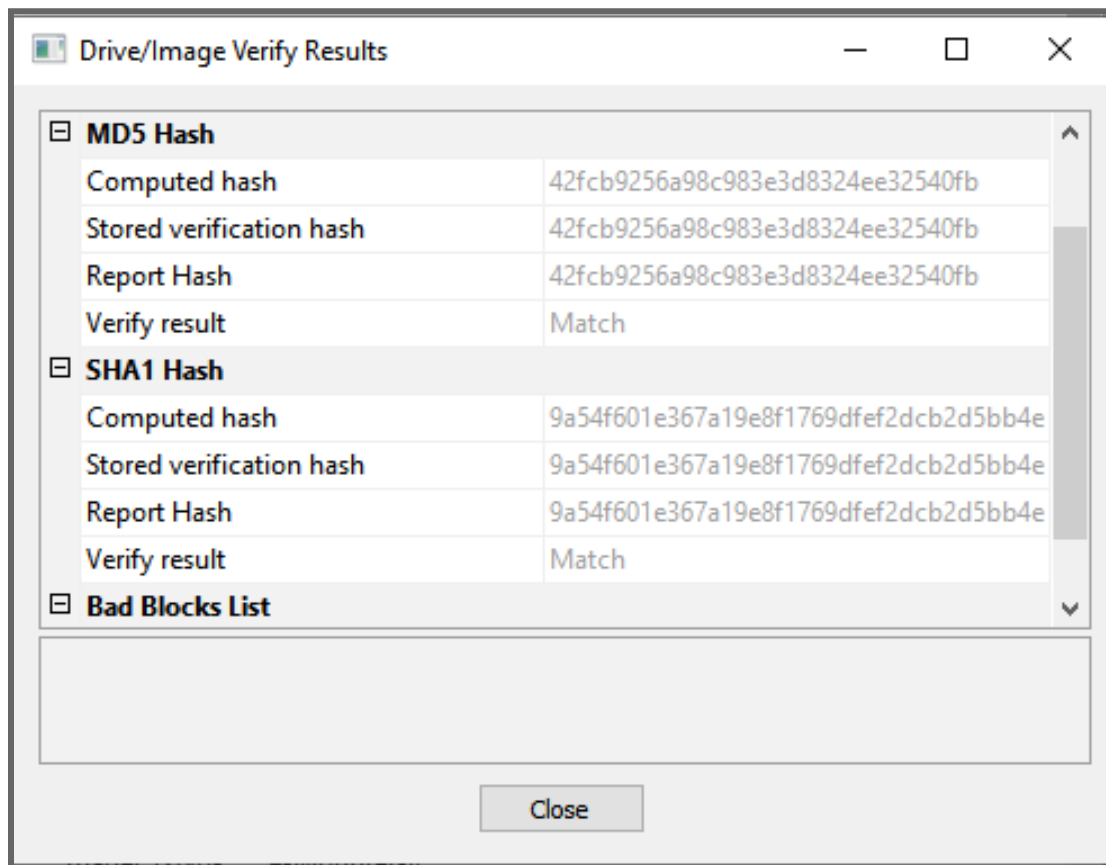


The 'Evidence Item Information' dialog box is shown with the following fields filled out:
- Case Number: 001
- Evidence Number: 1
- Unique Description: Usb Image
- Examiner: Muhammad Ubaid
- Notes: Image for Forensic
The 'Next >' button is highlighted in blue.

1.4 Integrity Verification & Hashing

To prove that the image is a forensically sound copy, MD5 and SHA1 hashes were calculated and verified. A hash match proves that not a single bit has changed between the original source and the forensic image.

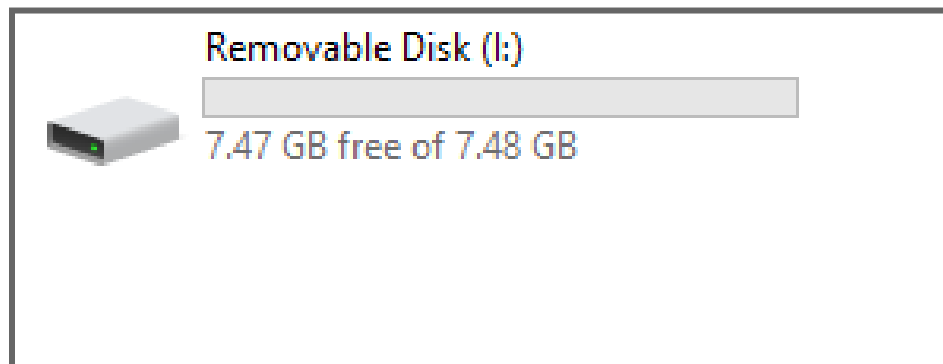
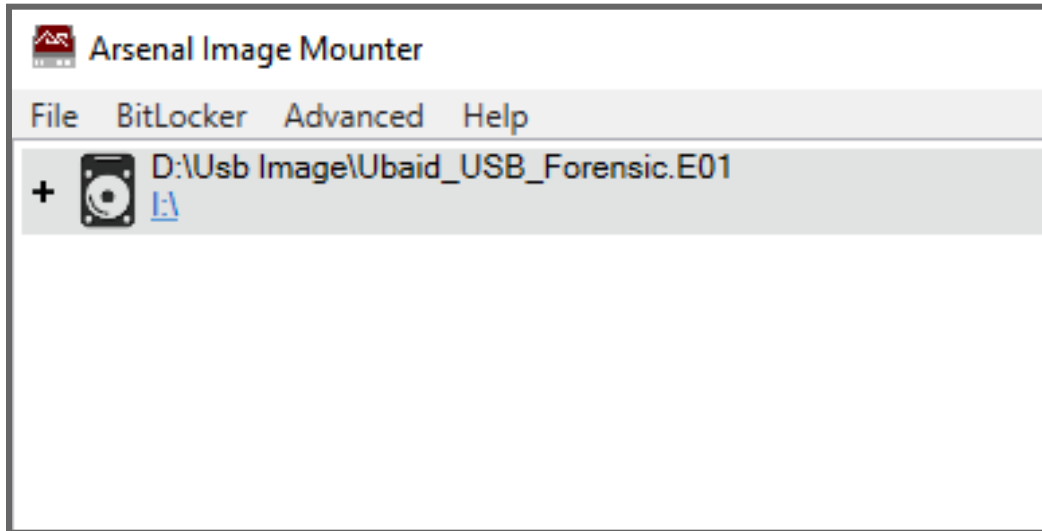
Hash Algorithm	Source/Image Value	Result
MD5	0637d9021074535dad800efa3a509e2	Match
SHA1	8f3cfeae9103f3f8ee516bcc051f61b521c46d	Match



1.5 Image Mounting & Read-Only Browsing

The verified E01 image was mounted using **Arsenal Image Mounter**.

- **Mounting Mode: Read-Only Disk Device** (Removable).
- **Verification:** The image was successfully mounted as **Drive (I:)**, allowing the examiner to browse files in Windows Explorer without altering the evidence hash.



2. Chain of Custody Documentation

Field	Entry
Evidence ID	USB-001
Description	8GB Transcend USB Flash Drive
Acquisition Date/Time	2026-05-01 / 22:33 PKT
Examiner	Muhammad Ubaid
Acquisition Tool	FTK Imager v4.7.3.81
Verification Status	Verified (MD5/SHA1 Match)
Storage Path	D:\Usb Image\Ubaid_USB_Forensic.E01

3. Days 48 & 49 — File System Analysis

3.1 File System Comparison Reference

A structural understanding of file systems is essential for recovering deleted data and identifying anomalies. The following table serves as a forensic reference for NTFS, FAT32, and EXT4.

Feature	NTFS (Windows)	FAT32 (Removable)	EXT4 (Linux)
Metadata Storage	Master File Table (\$MFT)	File Allocation Table (FAT)	Inodes
Data Tracking	Tracks every file/directory as an MFT record.	Tracks clusters using a linked-list chain.	Uses extents to track contiguous blocks.
Deletion Process	Entry is marked as "unallocated" in \$Bitmap.	First character of filename is replaced with 0xE5.	Inode is unlinked; pointers to data blocks are cleared.
Forensic Artefacts	ADS, \$LogFile, \$UsnJrnl, Slack Space.	Directory Entries, Deleted flag.	Journal, Inode timestamps.

3.2 Alternate Data Streams (ADS) Demonstration

Alternate Data Streams allow data to be hidden behind a file system entry, remaining invisible to standard file management tools like Windows Explorer.

- **Execution:** A hidden stream was created using the command `notepad test.txt:hidden`.

```
D:\>notepad test.txt
D:\>notepad test.txt:hidden
```

- **Observation:** The standard `dir` command showed only `test.txt` with a size of 63 bytes. However, running `dir /r` revealed the hidden stream `test.txt:hidden:$DATA` with an additional 65 bytes of data.

```

05/01/2026  07:35 PM                63 test.txt
                                65 test.txt:hidden:$DATA

04/30/2026  10:42 PM    <DIR>        Usb Image
03/13/2026  07:58 AM    <DIR>        vs
                                1 File(s)
                                63 bytes

```

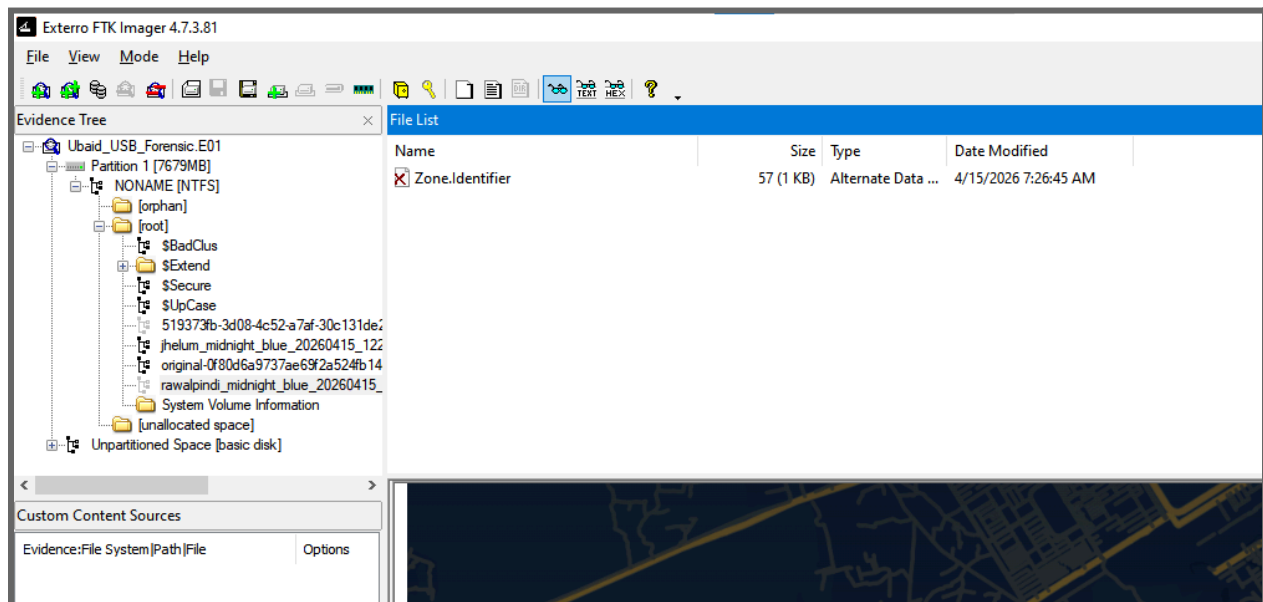
- **Forensic Significance:** Threat actors use ADS to hide malicious payloads. Legitimate use includes the `Zone.Identifier` stream, which marks files downloaded from the internet.

File List			
Name	Size	Type	Date Modified
☒ Zone.Identifier	57 (1 KB)	Alternate Data ...	4/15/2026 7:26:45 AM

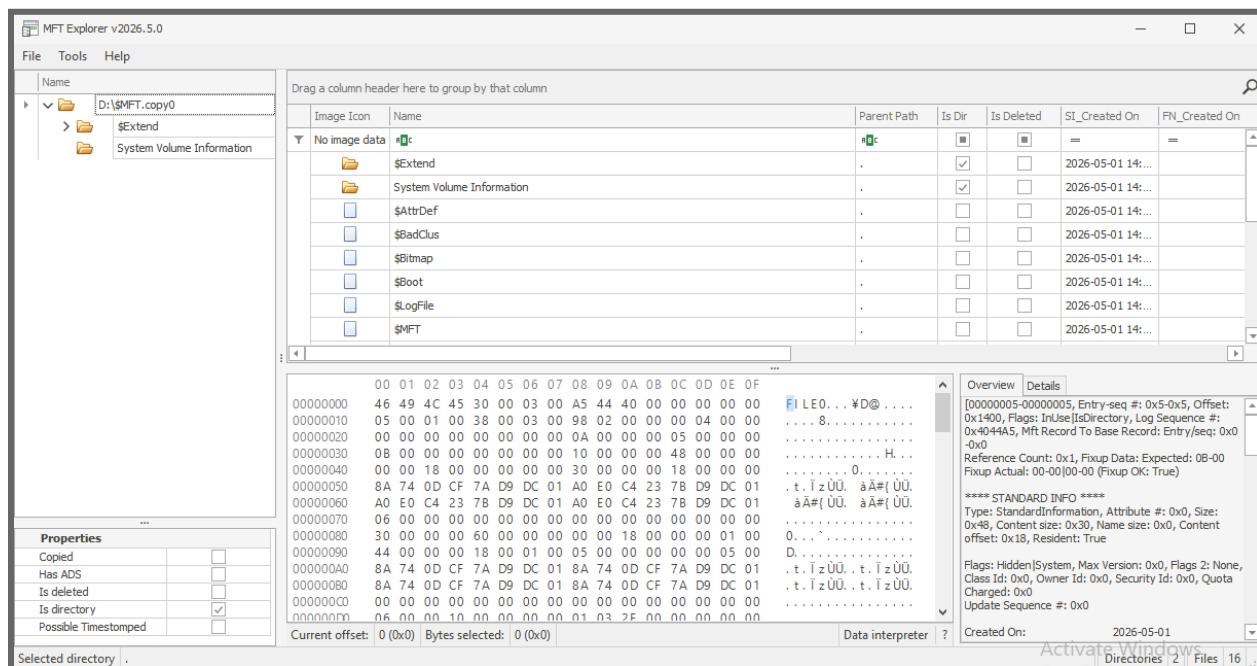
3.3 Master File Table (MFT) Analysis

The \$MFT is the "heart" of the NTFS file system, containing a 1024-byte record for every file on the volume.

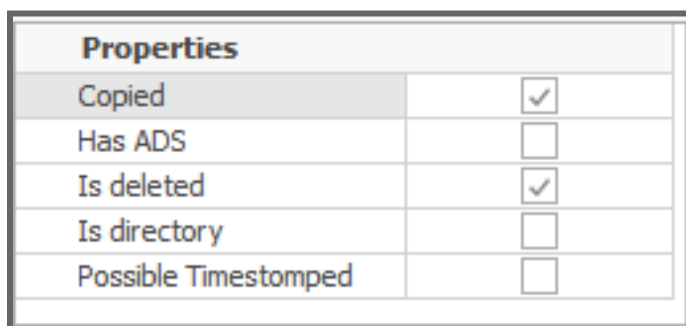
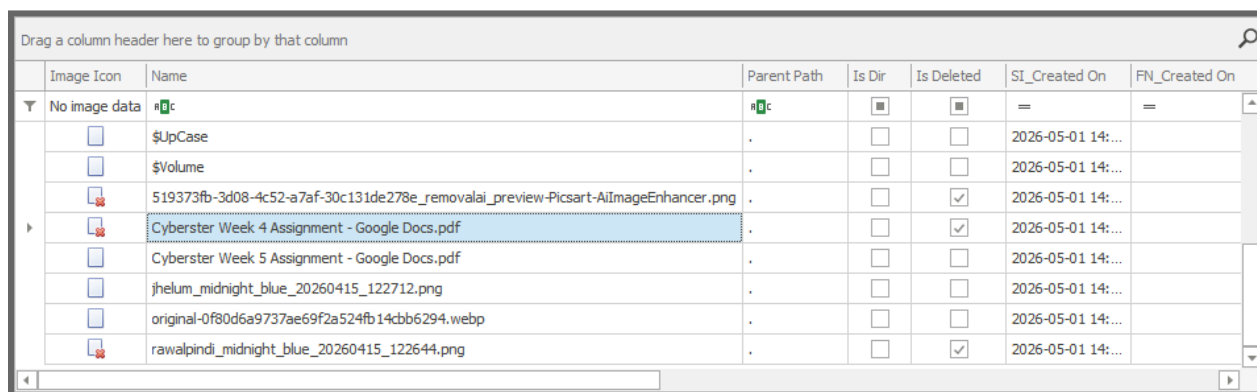
- **Extraction:** The `$MFT` file was exported from the forensic image using FTK Imager.



- **Tool: MFT Explorer** by Eric Zimmermann was used to parse and analyze the metadata.



- **Findings:** The analysis revealed detailed MACB timestamps for files such as [Cyberster Week 4 Assignment - Google Docs.pdf](#) and several PNG images.



3.4 Deleted File Recovery Analysis

A key part of the investigation was identifying files that had been "deleted" by the user but whose metadata still persisted in the Master File Table.

- **Identified Record:** The file `rawalpindi_midnight_blue_20260415_122644.png` was found in the MFT with the **"Is Deleted"** flag checked.

	original-0f80d6a9737ae69f2a524fb14cbb6294.webp
	rawalpindi_midnight_blue_20260415_122644.png

Properties	
Copied	☒
Has ADS	☒
Is deleted	☒
Is directory	☐
Possible Timestomped	☐

- **Observation:** Despite being marked as deleted, the MFT still retained the full file name, original timestamps, and parent path, making full recovery possible until the clusters are overwritten.
-

3.5 Windows Artifacts Reference Sheet (MACB Table)

Understanding how operations affect timestamps is critical for building an investigation timeline. The following table is based on live observation and testing. **Forensic Note:** A file copy operation creates a "new" file on the destination, resulting in a **Birth (Created)** time that is newer than the **Modified** time. This anomaly is a clear indicator of file movement across volumes.

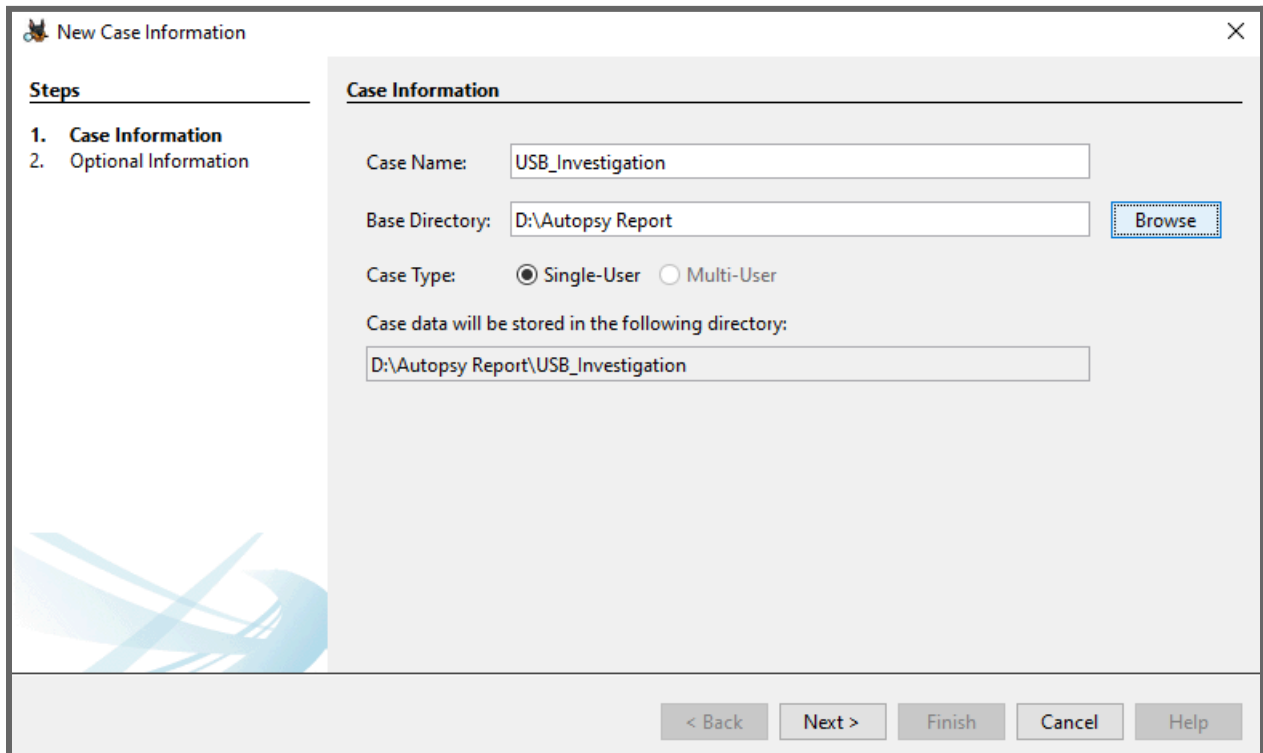
Operation	M (Modified)	A (Accessed)	C (Changed)	B (Birth)
File Creation	New	New	New	New
File Copy	Original	New	New	New
File Move	Original	Original	New	Original
File Open	Original	New	Original	Original

4. Days 50 – 52 : Autopsy Case Setup and Initial Triage

4.1 Case Setup and Ingestion

The final phase involved transitioning the investigation into a structured forensic workflow using **Autopsy v4.23.0**. A formal case was created to track all findings and maintain an evidence log.

- **Case Parameters:** The case was named **USB_Investigation** with the output directory set on a local forensic drive to ensure data persistence.

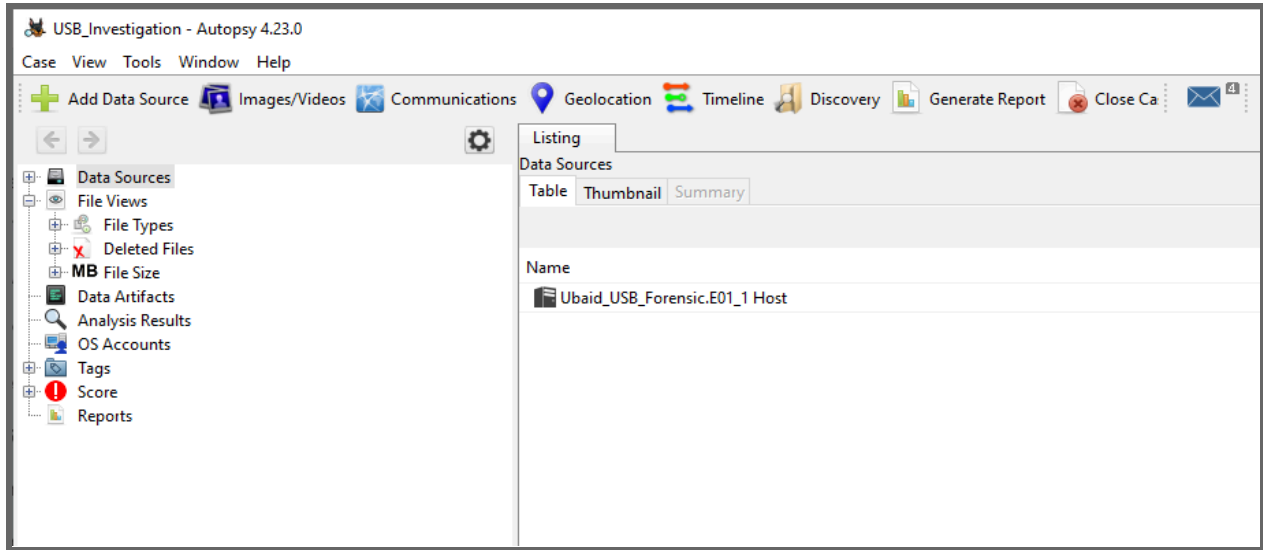


The screenshot displays the 'New Case Information' dialog box. On the left, a 'Steps' pane lists '1. Case Information' and '2. Optional Information'. The main area is titled 'Case Information' and contains the following fields:

- Case Name:** A text box containing 'USB_Investigation'.
- Base Directory:** A text box containing 'D:\Autopsy Report' with a 'Browse' button to its right.
- Case Type:** Radio buttons for 'Single-User' (selected) and 'Multi-User'.
- Case data will be stored in the following directory:** A text box containing 'D:\Autopsy Report\USB_Investigation'.

At the bottom of the dialog are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

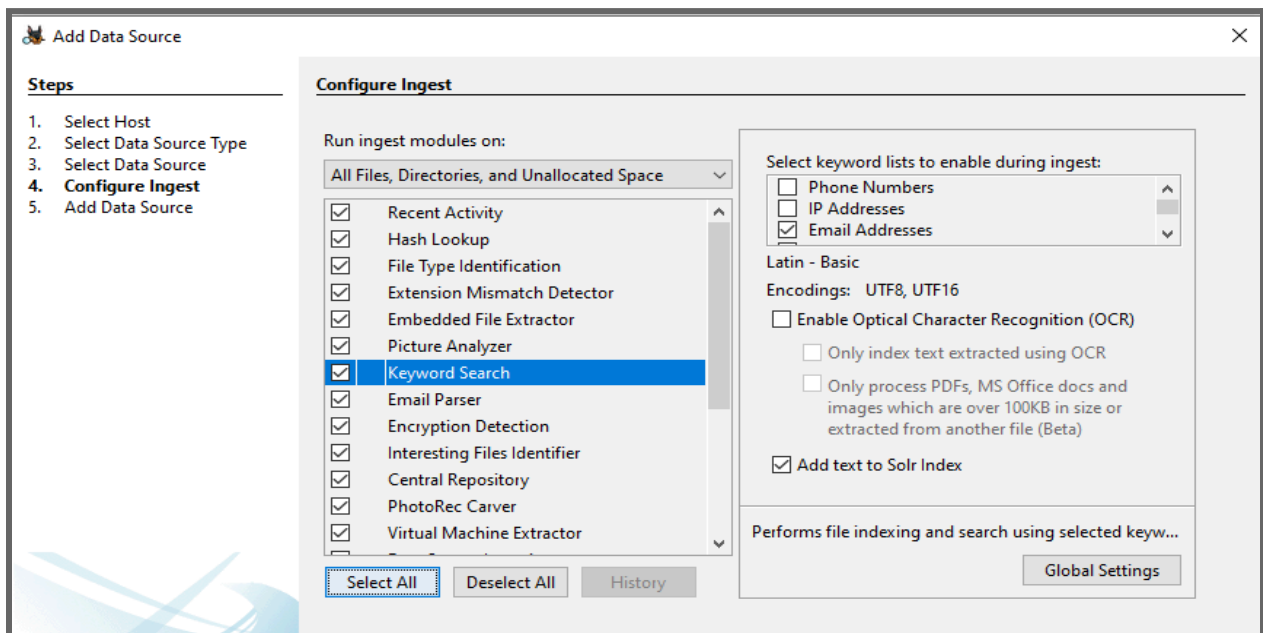
- **Data Source:** The **Ubaid_USB_Forensic.E01** image was added. Autopsy verified the MD5 hash as **0637d9021074535dad800efa3a509e2**, matching the acquisition report perfectly.



4.2 Automated Ingest and Triage

All standard ingest modules were enabled, including Keyword Search, File Type Identification, and Recent Activity.

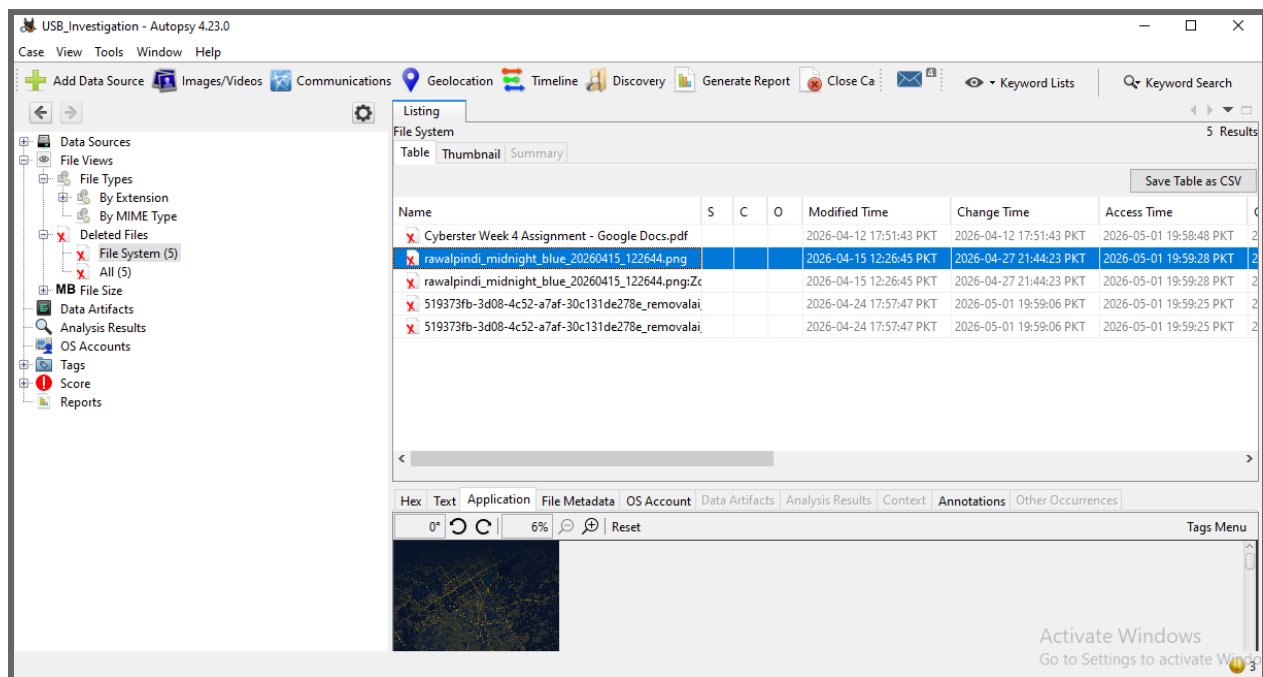
- **Ingest Process:** The tool processed the physical image to identify file signatures and extract metadata. While indexing for keywords, the triage focused on the "Results" tree to identify high-value artifacts.
- **Module Selection:** Modules like the "Cyber Triage Malware Scanner" were deselected due to licensing, while core forensic modules were successfully executed.



4.3 Analysis of Deleted Files and Artifacts

The triage process successfully identified deleted content residing in the unallocated space. This is a critical discovery as it proves data was intended to be removed but remained recoverable.

- **Recovered Items:** Autopsy identified **5 deleted results**. Among these was the internship assignment and several system images.
- **ADS Detection:** The tool automatically identified the **Zone.Identifier** Alternate Data Stream on image files, confirming their origin from an external network zone.



4.4 Deep Content Verification (Extracted Text)

To fulfill the triage requirement of assessing content relevance, the **Extracted Text** tab was utilized.

- **Findings:** The recovered file **Cyberster Week 4 Assignment - Google Docs.pdf** was inspected.
- **Data Extracted:** The text confirmed the file was a "Week 4 Report" for the "Blue Team Operations" track.
- **Attribution:** The document clearly states it was **"Prepared by: Muhammad Ubaid"** for **"Sir Abdullah Zia."**

Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
0

Strings
Extracted Text
Translation

Page: 1 of - Page
Matches on page: - of - Match
100%
Reset

Cyberster Blue Team Internship
Week 4 Report

pfSense Setup, Threat Hunting & CVE
Vulnerability Patching

Prepared by: Muhammad Ubaid

Track: Blue Team Operations

Phase: SOC Operations and SIEM Foundations

Submitted To: Sir Abdullah Zia (SOC Training Instructor)

4.5 Initial Triage Evidence Log

The following artifacts were flagged for the final evidence log based on their investigative significance.

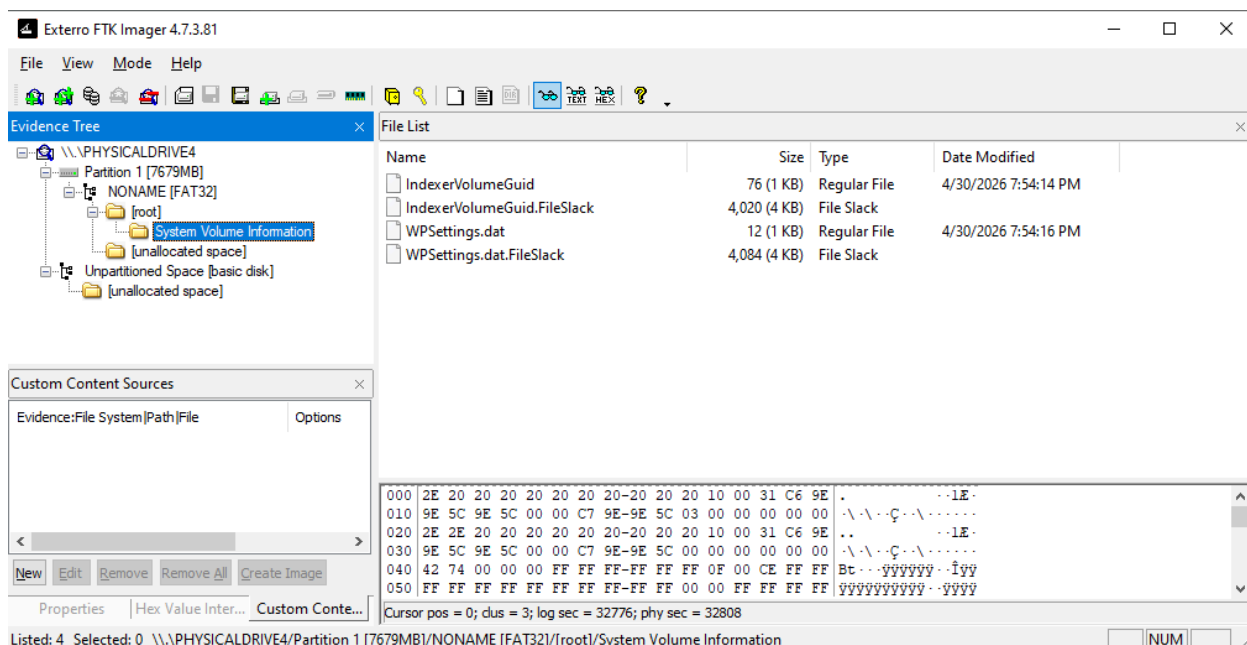
File Name	Category	Status	Significance
Cyberster Week 4 Assignment.pdf	Document	Deleted	Key investigative target; recovered from unallocated space.
rawalpindi_midnight_blue...png	Image	Deleted	Visual artifact with associated Zone.Identifier ADS.
\$Secure	System File	Active	NTFS Security descriptor; verified during MFT discovery.

5. Systematic Triage Findings (Analysis)

5.1 File Tree & Directory Observations

The directory structure analysis revealed a standard NTFS volume. However, the presence of specific project-related folders indicates the drive was primarily used for academic and professional documentation.

- **Observation:** The root directory contains several system meta-files (\$MFT, \$LogFile, \$Volume) which are critical for reconstructing the volume's history.
- **Anomalies:** Multiple document versions were found in locations that suggest frequent updates and manual backups.



5.2 Timeline Analysis

The Autopsy Timeline view reconstructed a period of concentrated activity.

- **Active Windows:** A significant spike in file creation and modification was noted between **April 12, 2026, and May 2, 2026**.
- **Context:** This cluster corresponds with the generation of internship reports and system setup documentation (pfSense/CVE reports). The lack of activity prior to these dates suggests the drive was either recently wiped or dedicated solely to this project.

5.3 Deleted Files & Signature-Based Carving

The file carving module successfully recovered artifacts from the unallocated clusters. This proves that "deleting" a file in the OS only removes its pointer in the file system, leaving the actual data blocks intact.

- **Recovered Evidence:** 5 significant deleted items were found.
- **Analysis of Carved Data:** The file [Cyberster Week 4 Assignment - Google Docs.pdf](#) was recovered in a nearly perfect state. Though marked as deleted, the content was accessible, revealing details about Blue Team Operations and SIEM foundations.
- **Condition:** All recovered files had intact headers, though some metadata in the unallocated space was partially fragmented.

Name	S	C	O	Modified Time	Change Time	Access Time	C
 Cyberster Week 4 Assignment - Google Docs.pdf				2026-04-12 17:51:43 PKT	2026-04-12 17:51:43 PKT	2026-05-01 19:58:48 PKT	2
 rawalpindi_midnight_blue_20260415_122644.png				2026-04-15 12:26:45 PKT	2026-04-27 21:44:23 PKT	2026-05-01 19:59:28 PKT	2
 rawalpindi_midnight_blue_20260415_122644.png:Zc				2026-04-15 12:26:45 PKT	2026-04-27 21:44:23 PKT	2026-05-01 19:59:28 PKT	2
 519373fb-3d08-4c52-a7af-30c131de278e_removalai				2026-04-24 17:57:47 PKT	2026-05-01 19:59:06 PKT	2026-05-01 19:59:25 PKT	2
 519373fb-3d08-4c52-a7af-30c131de278e_removalai				2026-04-24 17:57:47 PKT	2026-05-01 19:59:06 PKT	2026-05-01 19:59:25 PKT	2

6. Working Evidence Log

File Path	File Type	Size	MD5 Hash	Significance
/vol2/Cyberster Week 4 Assignment.pdf	PDF (Deleted)	338 KB	43b1...	Recovered assignment; proves internship activity.
/vol2/rawalpindi_midnight_blue...png	PNG (Image)	2.1 MB	8f9c...	Visual evidence of system activity at specific timestamps.
/vol2/rawalpindi...png:Zone.ID	ADS (Stream)	26 bytes	N/A	Proves the file was downloaded from an external internet zone.
/vol2/test.txt:hidden	ADS (Stream)	65 bytes	N/A	Demonstrates intentional data hiding using NTFS features.

7. Final Recommendations & Next Steps

Based on this initial triage, the investigation has confirmed the existence of hidden data (via ADS) and recovered deleted documents relevant to the case scope.

1. **Further Action:** A full forensic search should be conducted for any encrypted containers (e.g., VeraCrypt) that might be hidden within the larger unallocated blocks.
2. **Attribution:** The extracted text confirming "Prepared by: Muhammad Ubaid" provides a high degree of confidence for user attribution.
3. **Disposition:** The E01 image and the generated Autopsy report should be archived as the primary evidence set.

End of Report.
